

PROGRAM 4 - Create a dockerized application and push it to GitHub

using Personal Access Token (PAT).

Steps

1. Create project directory

`mkdir lab4 && cd lab4`

2. Create files

- `app.py`
- `requirements.txt`
- `Dockerfile`

3.Create Dockerfile

Create Dockerfile:`nano Dockerfile`

`FROM python:latest`

`WORKDIR /app`

`COPY requirements.txt .`

`RUN pip install --no-cache-dir -r requirements.txt`

`COPY . .`

`EXPOSE 5000`

`CMD ["python", "app.py"]`

3. Build Docker image

`docker build -t program4 .`

4. Run Docker container

`docker run -p 5000:5000 program4`

5. Verify output

Open browser:<http://localhost:5000>

6. Initialize Git

`git init`

`git branch -M main`

git add .

git commit -m "initial commit"

7. Push to GitHub using PAT

git remote add origin https://github.com/<username>/program4.git

git push -u origin main

OUTPUT

Dockerized application created and pushed to GitHub successfully.

